MANIPAL INSTITUTE OF TECHNOLOGY

Manipal

(A constituent unit of MAHE, Manipal)

III SEMESTER B. TECH (CSFT)

Course name (Course code): Programming Paradigm Laboratory (CEF2111)

Lab Course Objectives and **Course Outcomes** aligned with the programming paradigms and experiments. These are crafted to reflect outcome-based education principles, Bloom's taxonomy, and your pedagogical clarity.

Lab Course Objectives (LCOs)

1. **To impart foundational and advanced understanding of programming paradigms** Enable students to explore procedural, object-oriented, and functional programming through hands-on experiments involving parameter passing, pointers, inheritance, and recursion.

2. **To develop modular and safe coding practices using Java and C constructs** Equip learners with the ability to implement type-safe, memory-safe, and concurrency-aware programs using pointers, collections, threads, and generics.

3. **To foster higher-order thinking through reflective and meta-programming techniques** Encourage students to inspect, abstract, and dynamically manipulate program behavior using reflection, currying, and closures for real-world applications like invoice generation and banking systems.

Course Outcomes (COs)

CO Code	Description	Bloom's Level
CO1	Apply parameter passing techniques and pointer types to design memory-safe modular programs.	Apply (L3)
CO2	Implement object-oriented and functional constructs including inheritance, collections, and closures to solve domain-specific problems.	Analyze (L4)
CO3	Demonstrate the ability to inspect, abstract, and dynamically control program behavior using recursion, reflection, and meta-programming.	Evaluate (L5)

1. **To impart foundational and advanced understanding of programming paradigms:** Enable students to explore procedural and functional programming through hands-on experiments involving parameter passing, pointers, and recursion.

2. **To develop modular and safe coding practices using** object-oriented programming: Equip learners with the ability to implement type-safe, memory-safe, and concurrency-aware programs using pointers, collections, threads, and generics.

3. **To foster higher-order thinking through reflective and meta-programming techniques and covariance and contravariance**: Encourage students to inspect, abstract, and dynamically manipulate program behavior using reflection, currying, and closures for real-world applications like invoice generation and banking systems.

LIST OF EXPERIMENTS

EXP 1 – Parameter Passing & Safety

EXP 2 – Types of pointers

Types of Pointers (with Examples)

a. Null Pointer
b. Void Pointer
c. Dangling Pointer
d. Pointer to Pointer
e. Wild Pointer
f. Constant Pointer
g. Pointer to Constant

EXP 3 – Function category types using arrays, pointers and structures

- Without Arguments and Without Return Value
- Without Arguments but With Return Value
- With Arguments but Without Return Value
- With Arguments and With Return Value

EXP 4 – Inheritance types in OOP (Single-Multilevel-Hierarchical inheritance, Hybrid and multiple inheritance through interfaces)

EXP 5 – Java Collection framework – linked list, array list, hash set, tree set, hash map, tree map

EXP 6 – Exploring Types of Recursions Head, Tail, and Tree, indirect Recursion types

EXP 7 – Class inspection using reflexive programming

EXP 8 – Meta programming features

EXP 9 – Creating threads in java using thread class and runnable interface, Thread class methods

EXP 10 – Covariance and Contravariance in java generics

EXP 11 – Dynamic Invoice Generator using Function Currying

EXP 12 – Bank Account System using Closures

THE END